Other Services

Overview of Services that might come up in a few questions

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What is CloudFormation



- CloudFormation is a declarative way of outlining your AWS Infrastructure, for any resources (most of them are supported).
- For example, within a CloudFormation template, you say:
  - I want a security group
  - I want two EC2 instances using this security group
  - I want an S3 bucket
  - I want a load balancer (ELB) in front of these machines
- Then CloudFormation creates those for you, in the **right order**, with the **exact configuration** that you specify

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Benefits of AWS CloudFormation (1/2)

- Infrastructure as code
  - No resources are manually created, which is excellent for control
  - Changes to the infrastructure are reviewed through code

- Cost
  - Each resources within the stack is tagged with an identifier so you can easily see how much a stack costs you
  - You can estimate the costs of your resources using the CloudFormation template
  - Savings strategy: In Dev, you could automation deletion of templates at 5 PM and recreated at 8 AM, safely

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Benefits of AWS CloudFormation (2/2)

- Productivity
  - Ability to destroy and re-create an infrastructure on the cloud on the fly
  - Automated generation of Diagram for your templates!
  - Declarative programming (no need to figure out ordering and orchestration)

- Don't re-invent the wheel
  - Leverage existing templates on the web!
  - Leverage the documentation

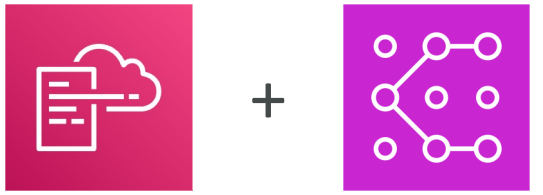
- Supports (almost) all AWS resources:
  - Everything we'll see in this course is supported
  - You can use "custom resources" for resources that are not supported

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CloudFormation + Infrastructure Composer

- Example: WordPress CloudFormation Stack



- We can see all the **resources**
- We can see the **relations** between the components

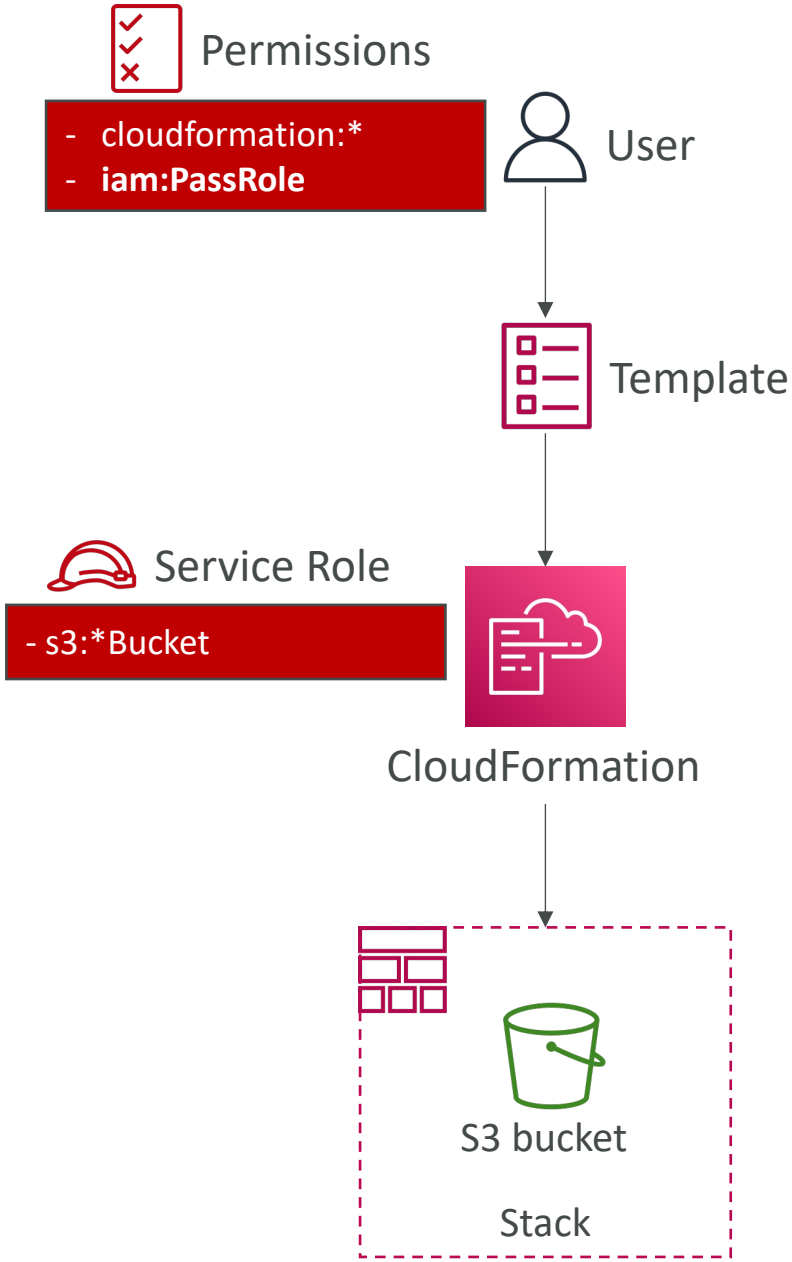


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CloudFormation – Service Role

- IAM role that allows CloudFormation to create/update/delete stack resources on your behalf
- Give ability to users to create/update/delete the stack resources even if they don't have permissions to work with the resources in the stack
- Use cases:
  - You want to achieve the least privilege principle
  - But you don't want to give the user all the required permissions to create the stack resources
- User must have **iam:PassRole** permissions



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Amazon Simple Email Service (Amazon SES)



- **Fully managed service to send emails securely, globally and at scale**
- Allows inbound/outbound emails
- Reputation dashboard, performance insights, anti-spam feedback
- Provides statistics such as email deliveries, bounces, feedback loop results, email open
- Supports DomainKeys Identified Mail (DKIM) and Sender Policy Framework (SPF)
- Flexible IP deployment: shared, dedicated, and customer-owned IPs
- Send emails using your application using AWS Console, APIs, or SMTP
- Use cases: transactional, marketing and bulk email communications

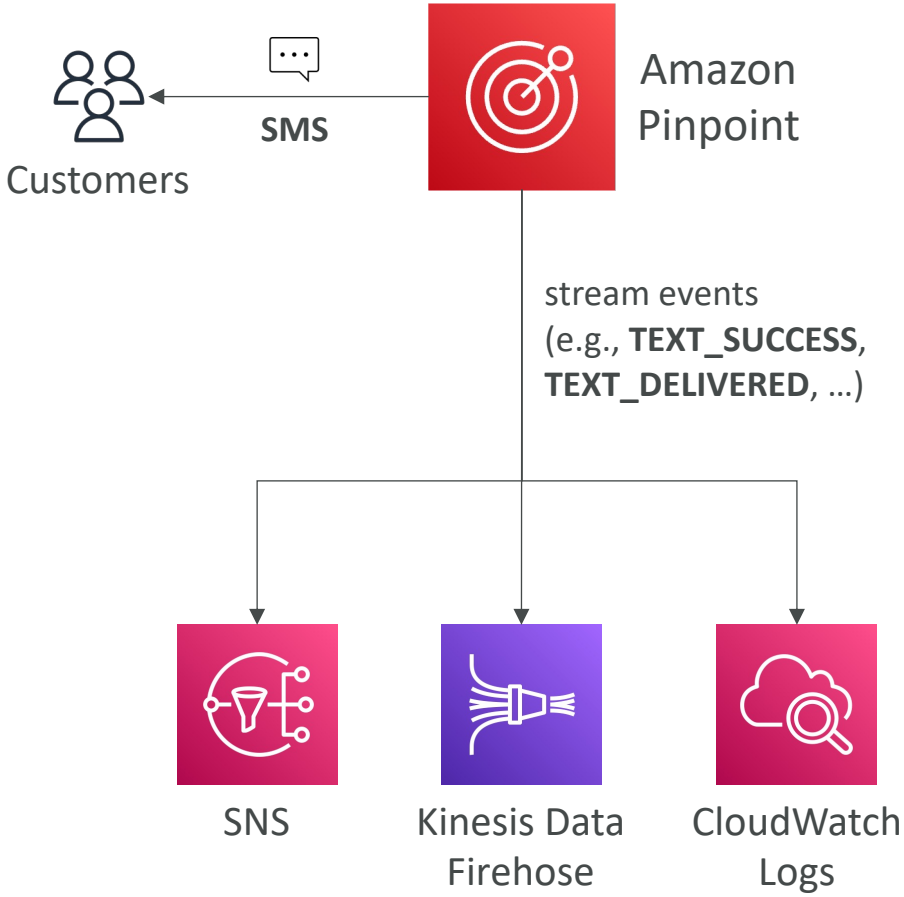


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Amazon Pinpoint

- Scalable **2-way (outbound/inbound)** marketing communications service
- Supports email, SMS, push, voice, and in-app messaging
- Ability to segment and personalize messages with the right content to customers
- Possibility to receive replies
- Scales to billions of messages per day
- Use cases: run campaigns by sending marketing, bulk, transactional SMS messages
- **Versus Amazon SNS or Amazon SES**
  - In SNS & SES you managed each message's audience, content, and delivery schedule
  - In Amazon Pinpoint, you create message templates, delivery schedules, highly-targeted segments, and full campaigns



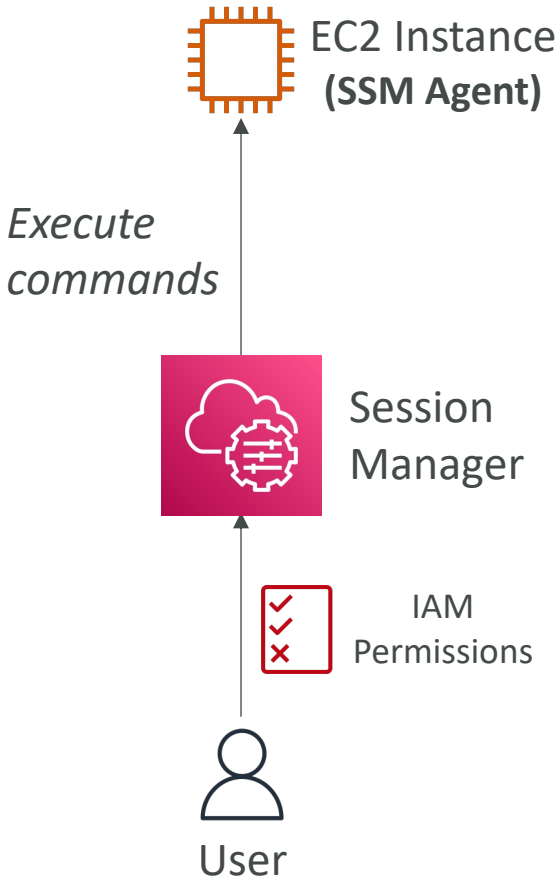
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Systems Manager – SSM Session Manager

- Allows you to start a secure shell on your EC2 and on-premises servers

- No SSH access, bastion hosts, or SSH keys needed
- No port 22 needed (better security)
- Supports Linux, macOS, and Windows
- Send session log data to S3 or CloudWatch Logs

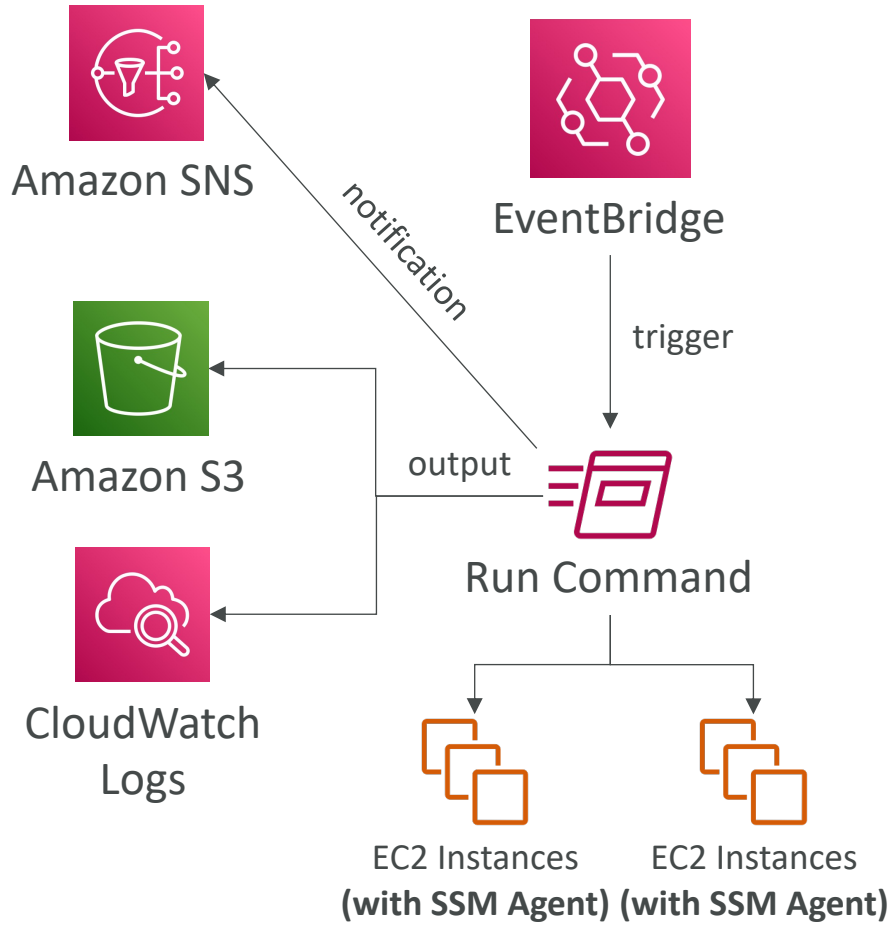


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Systems Manager – Run Command

- Execute a document (= script) or just run a command
- Run command across multiple instances (using resource groups)
- No need for SSH
- Command Output can be shown in the AWS Console, sent to S3 bucket or CloudWatch Logs
- Send notifications to SNS about command status (In progress, Success, Failed, …)
- Integrated with IAM & CloudTrail
- Can be invoked using EventBridge

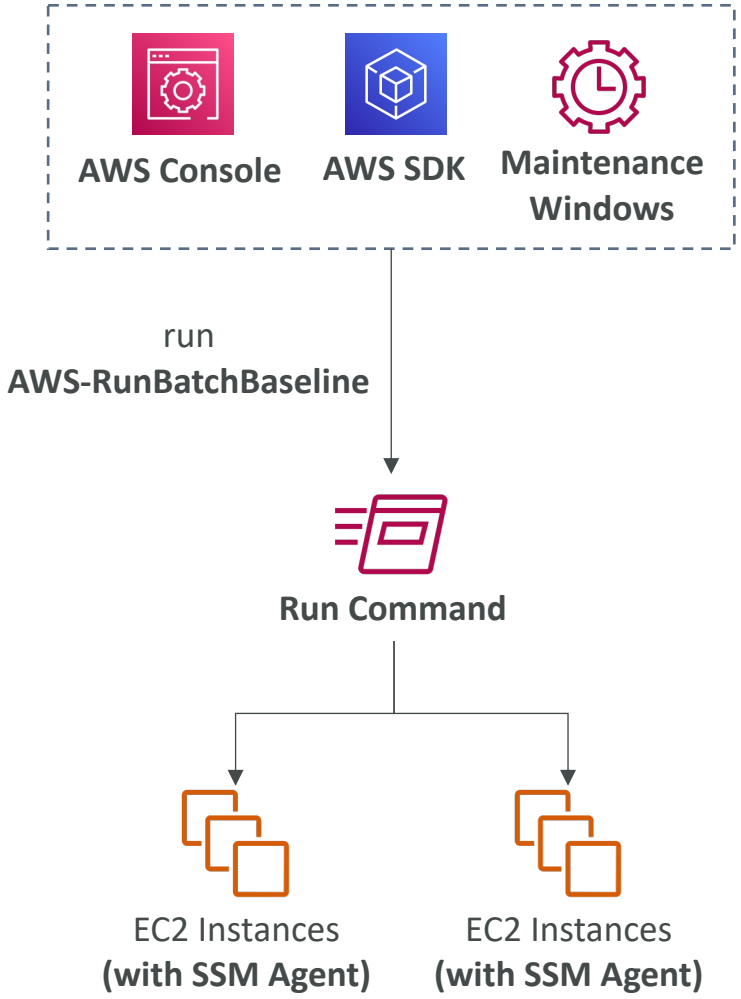


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Systems Manager – Patch Manager

- Automates the process of patching managed instances
- OS updates, applications updates, security updates
- Supports EC2 instances and on-premises servers
- Supports Linux, macOS, and Windows
- Patch on-demand or on a schedule using **Maintenance Windows**
- Scan instances and generate patch compliance report (missing patches)

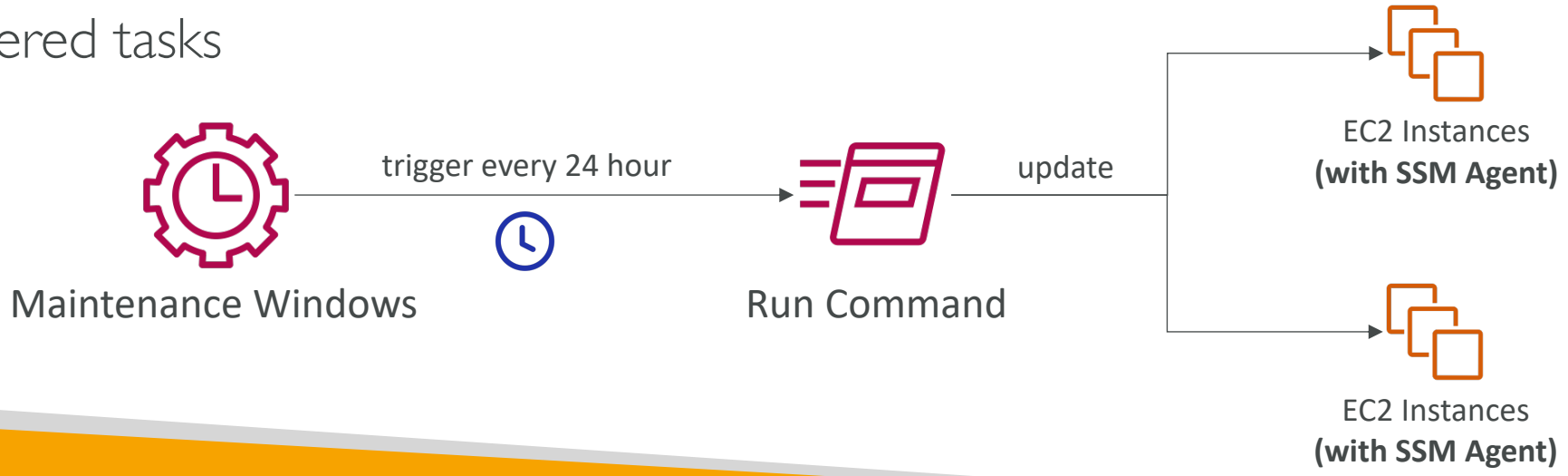


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Systems Manager – Maintenance Windows

- Defines a schedule for when to perform actions on your instances
- Example: OS patching, updating drivers, installing software, …
- Maintenance Window contains
  - Schedule
  - Duration
  - Set of registered instances
  - Set of registered tasks

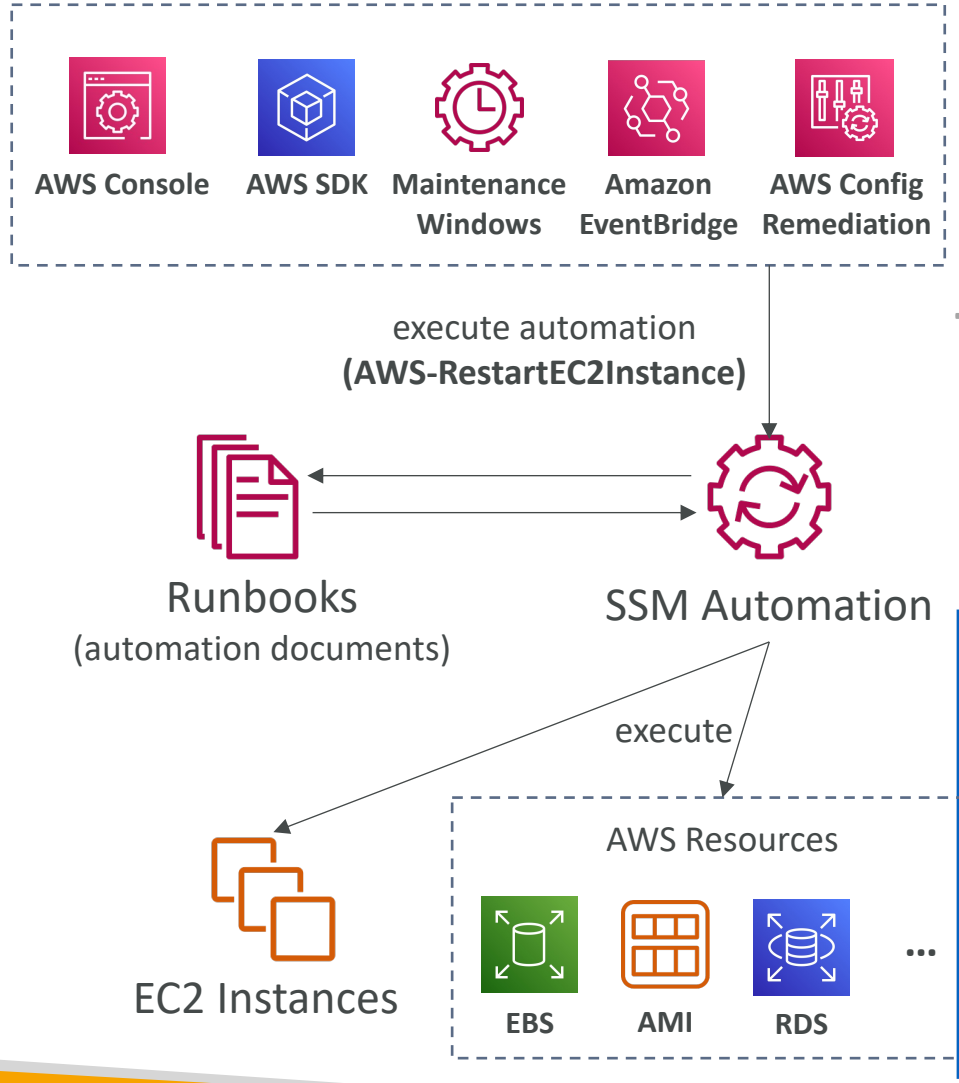


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Systems Manager - Automation

- Simplifies common maintenance and deployment tasks of EC2 instances and other AWS resources
- Examples: restart instances, create an AMI, EBS snapshot
- **Automation Runbook** – SSM Documents to define actions preformed on your EC2 instances or AWS resources (pre-defined or custom)
- Can be triggered using:
  - Manually using AWS Console, AWS CLI or SDK
  - Amazon EventBridge
  - On a schedule using Maintenance Windows
  - By AWS Config for rules remediations



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Cost Explorer



- Visualize, understand, and manage your AWS costs and usage over time
- Create custom reports that analyze cost and usage data.
- Analyze your data at a high level: total costs and usage across all accounts
- Or Monthly, hourly, resource level granularity
- Choose an optimal **Savings Plan** (to lower prices on your bill)
- **Forecast usage up to 18 months based on previous usage**

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Cost Explorer – Monthly Cost by AWS Service

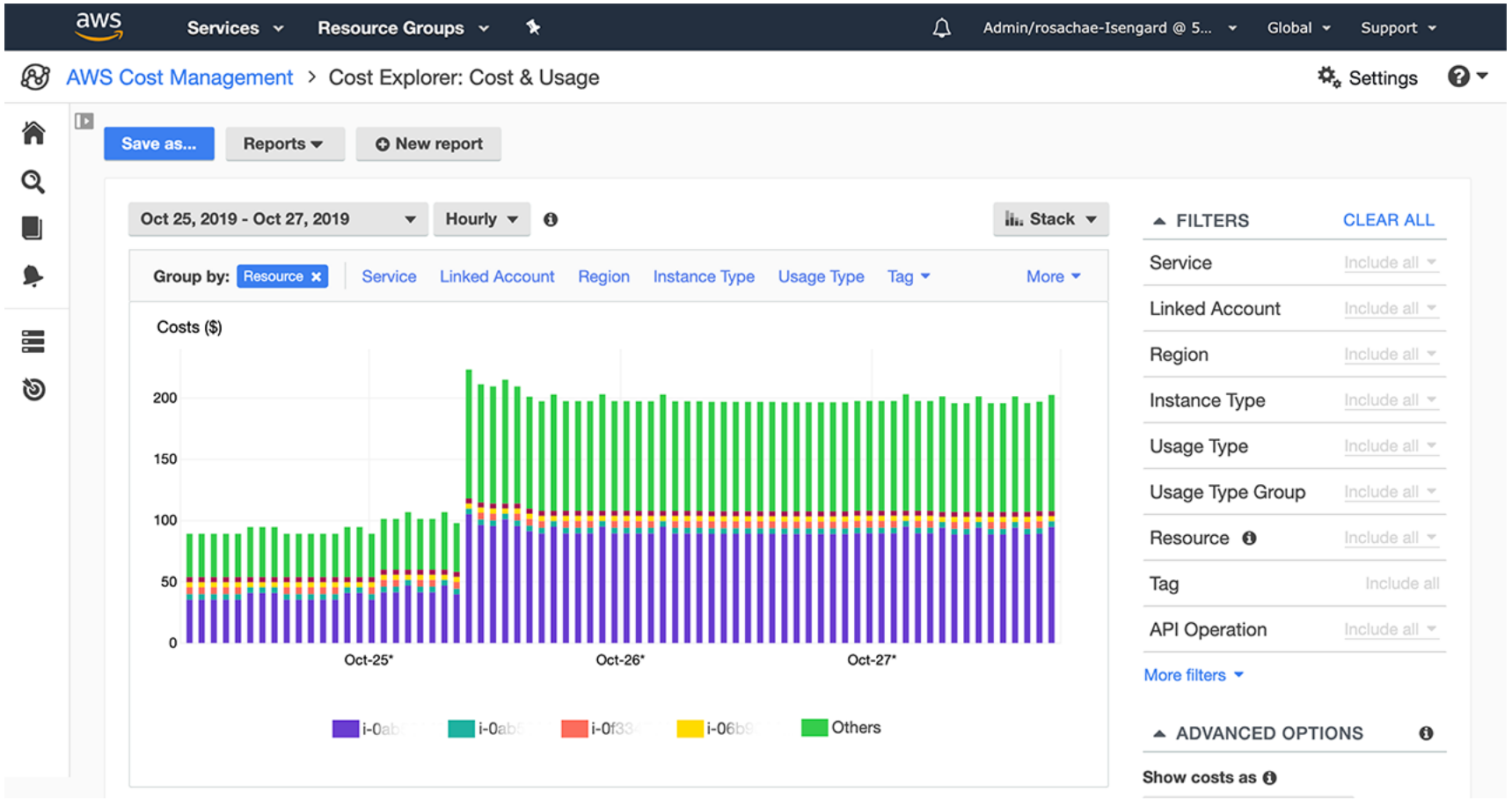


Instance Type	Oct 1, 2018	Nov 1, 2018	Dec 1, 2018	Jan 1, 2019
Total cost ($)	1,312.71	1,328.54	1,125.99	1,129.65
t2.micro ($)	486.75	475.89	405.63	409.27
c4.2xlarge ($)	296.11	286.56	296.11	296.11

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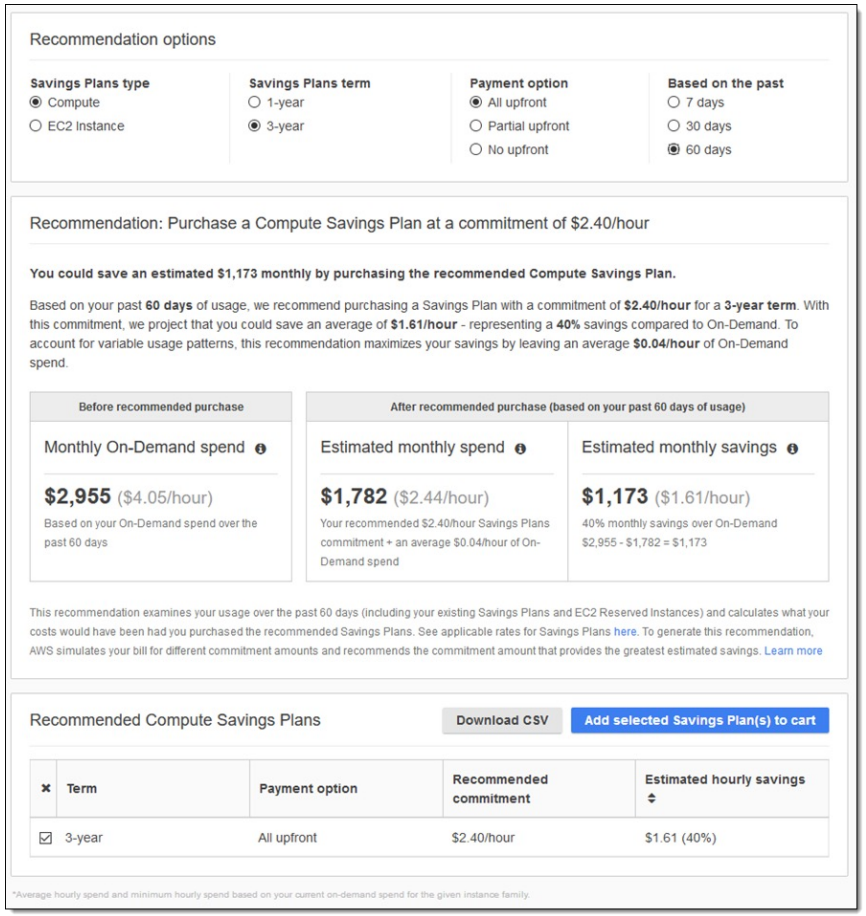
Cost Explorer– Hourly & Resource Level



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Cost Explorer – Savings Plan Alternative to Reserved Instances



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Cost Explorer – Forecast Usage

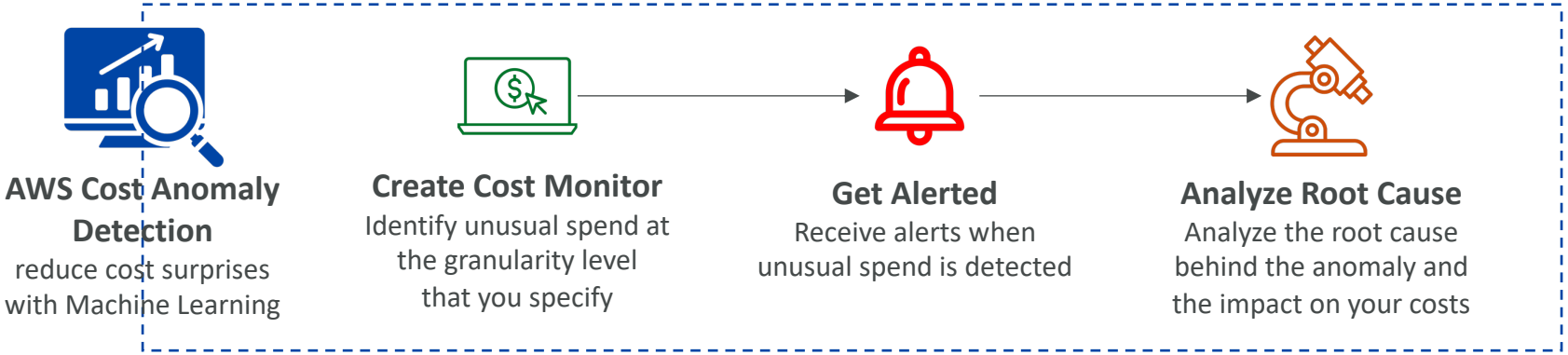


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AWS Cost Anomaly Detection

- Continuously monitor your cost and usage using ML to detect unusual spends
- It learns your unique, historic spend patterns to detect one-time cost spike and/or continuous cost increases (you don't need to define thresholds)
- Monitor AWS services, member accounts, cost allocation tags, or cost categories
- Sends you the anomaly detection report with root-cause analysis
- Get notified with individual alerts or daily/weekly summary (using SNS)

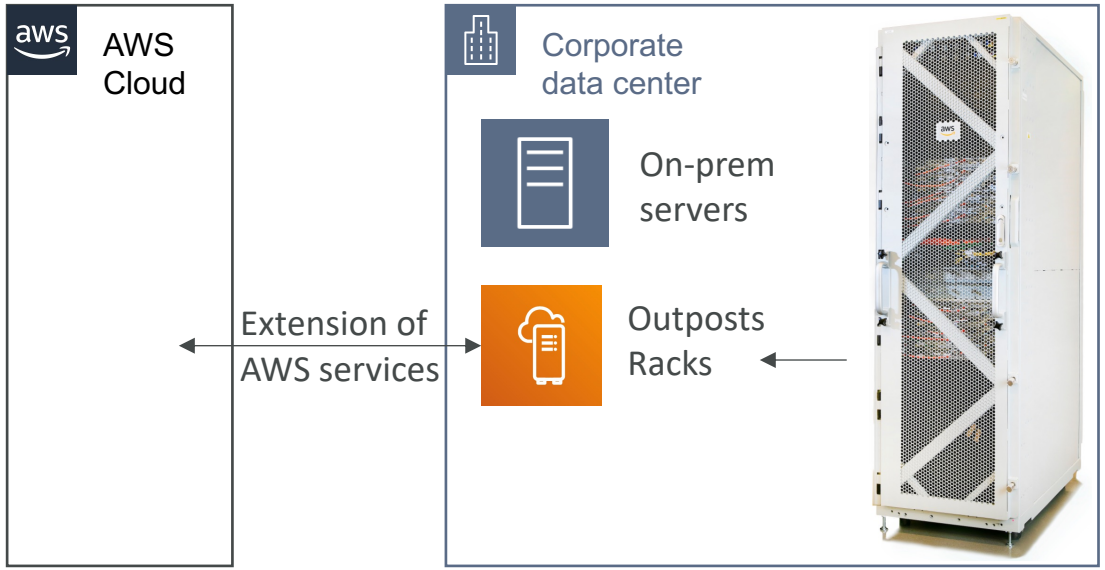


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AWS Outposts

- **Hybrid Cloud**: businesses that keep an on-premises infrastructure alongside a cloud infrastructure
- Therefore, two ways of dealing with IT systems:
  - One for the AWS cloud (using the AWS console, CLI, and AWS APIs)
  - One for their on-premises infrastructure
- **AWS Outposts are "server racks"** that offers the same AWS infrastructure, services, APIs & tools to build your own applications on-premises just as in the cloud
- **AWS will setup and manage "Outposts Racks"** within your on-premises infrastructure and you can start leveraging AWS services on-premises
- You are responsible for the Outposts Rack physical security



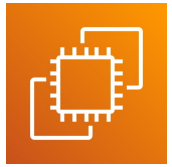
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AWS Outposts



- Benefits:
  - Low-latency access to on-premises systems
  - Local data processing
  - Data residency
  - Easier migration from on-premises to the cloud
  - Fully managed service
- Some services that work on Outposts:


Amazon EC2


Amazon EBS


Amazon S3


Amazon EKS


Amazon ECS


Amazon RDS


Amazon EMR

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AWS Batch

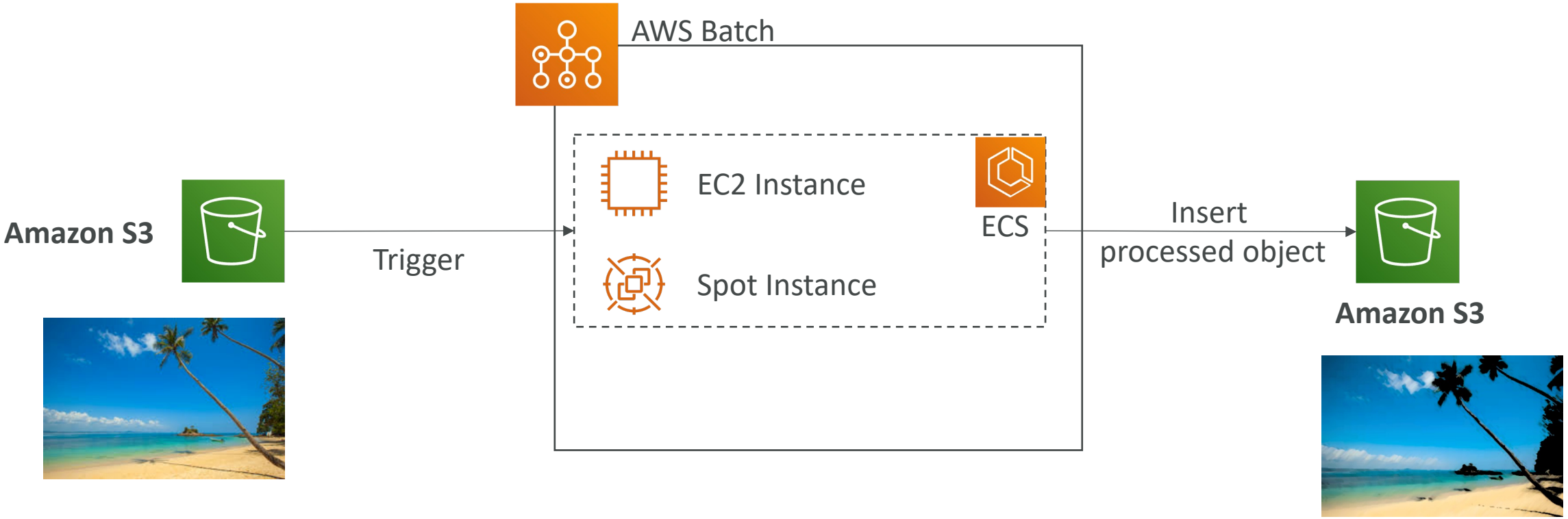


- **Fully managed** batch processing at **any scale**
- Efficiently run 100,000s of computing batch jobs on AWS
- A "batch" job is a job with a start and an end (opposed to continuous)
- Batch will dynamically launch **EC2 instances** or **Spot Instances**
- AWS Batch provisions the right amount of compute / memory
- You submit or schedule batch jobs and AWS Batch does the rest!
- Batch jobs are defined as **Docker images** and **run on ECS, EKS, Fargate**
- Helpful for cost optimizations and focusing less on the infrastructure

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AWS Batch – Simplified Example



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Batch vs Lambda

- Lambda:
  - Time limit
  - Limited runtimes
  - Limited temporary disk space
  - Serverless



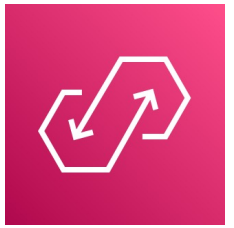
- Batch:
  - No time limit
  - Any runtime as long as it's packaged as a Docker image
  - Rely on EBS / instance store for disk space
  - Relies on EC2 (can be managed by AWS)



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Amazon AppFlow

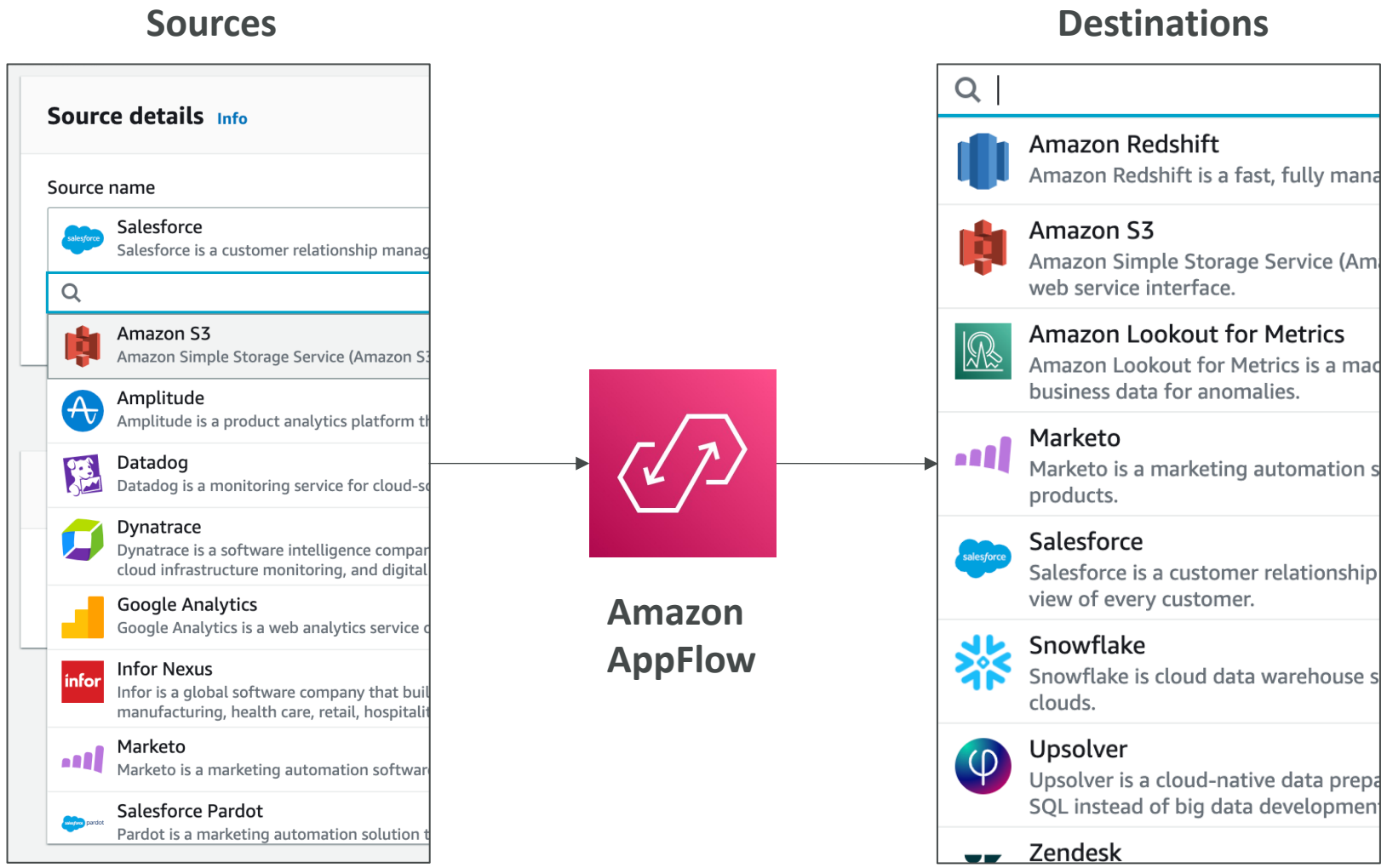


- Fully managed integration service that enables you to securely transfer data between **Software-as-a-Service (SaaS) applications and AWS**
- Sources: **Salesforce**, SAP, Zendesk, Slack, and ServiceNow
- Destinations: AWS services like **Amazon S3, Amazon Redshift** or non-AWS such as SnowFlake and Salesforce
- Frequency: on a schedule, in response to events, or on demand
- Data transformation capabilities like filtering and validation
- Encrypted over the public internet or privately over AWS PrivateLink
- Don't spend time writing the integrations and leverage APIs immediately

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Amazon AppFlow

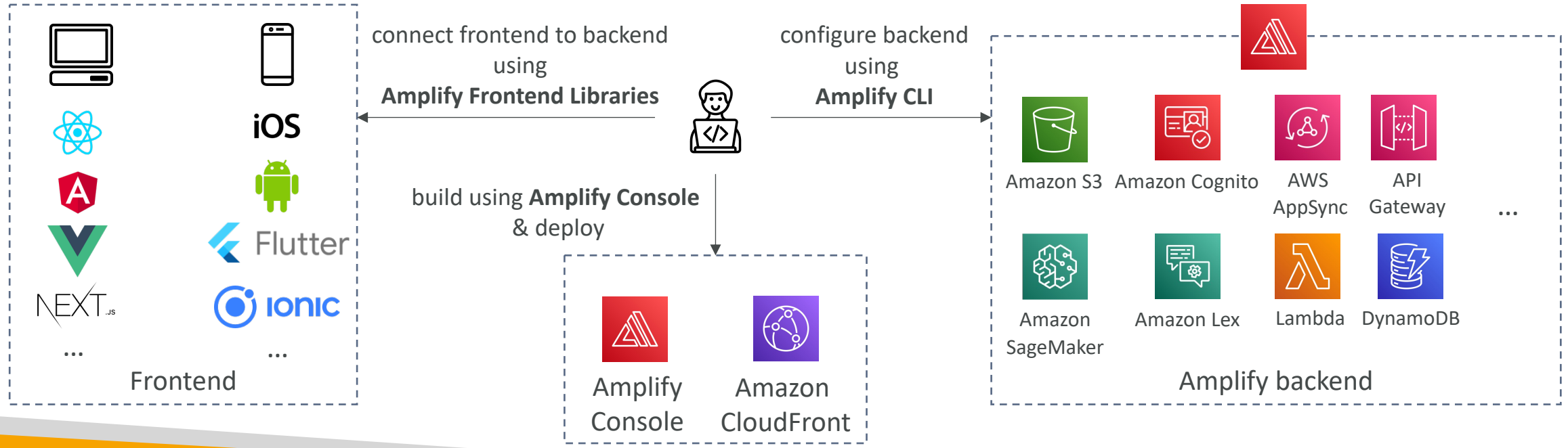


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AWS Amplify - web and mobile applications

- A set of tools and services that helps you develop and deploy scalable full stack web and mobile applications
- Authentication, Storage, API (REST, GraphQL), CI/CD, PubSub, Analytics, AI/ML Predictions, Monitoring, …
- Connect your source code from GitHub, AWS CodeCommit, Bitbucket, GitLab, or upload directly



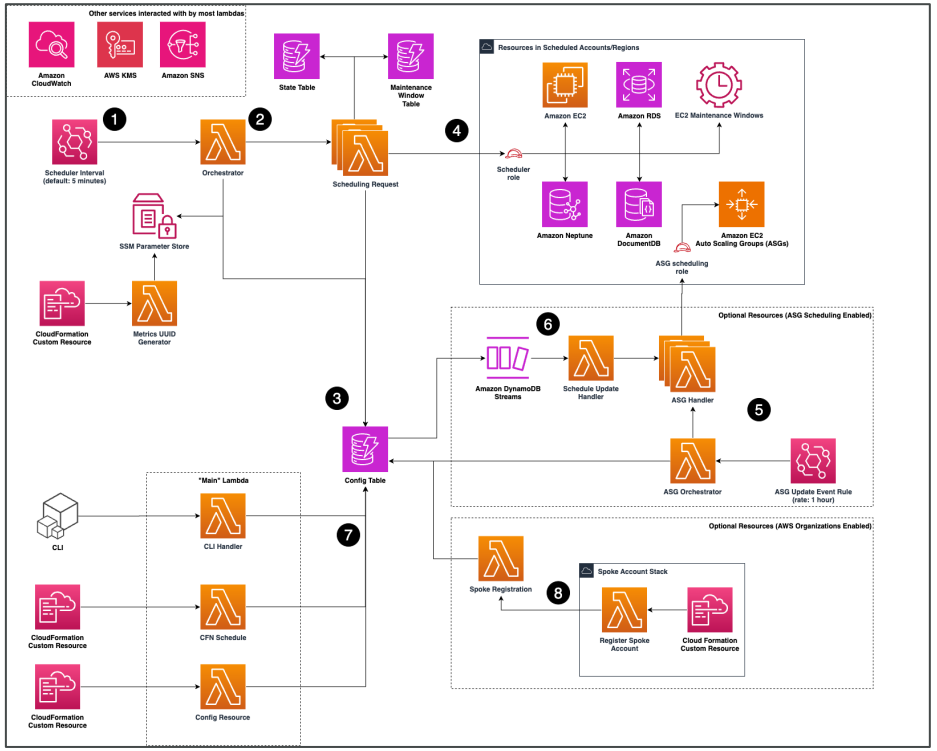
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Instance Scheduler on AWS

- AWS solution deployed through CloudFormation (not a service)
- **Automatically start/stop your AWS services to reduce costs (up to 70%)**
- Example: stop company's EC2 instances outside business hours
- **Supports EC2 instances, EC2 Auto Scaling Groups, and RDS instances**
- Schedules are managed in a DynamoDB table
- Uses resources' tags and Lambda to stop/start instances
- Supports cross-account and cross-region resources

- https://aws.amazon.com/solutions/implementations/instance-scheduler-on-aws/



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